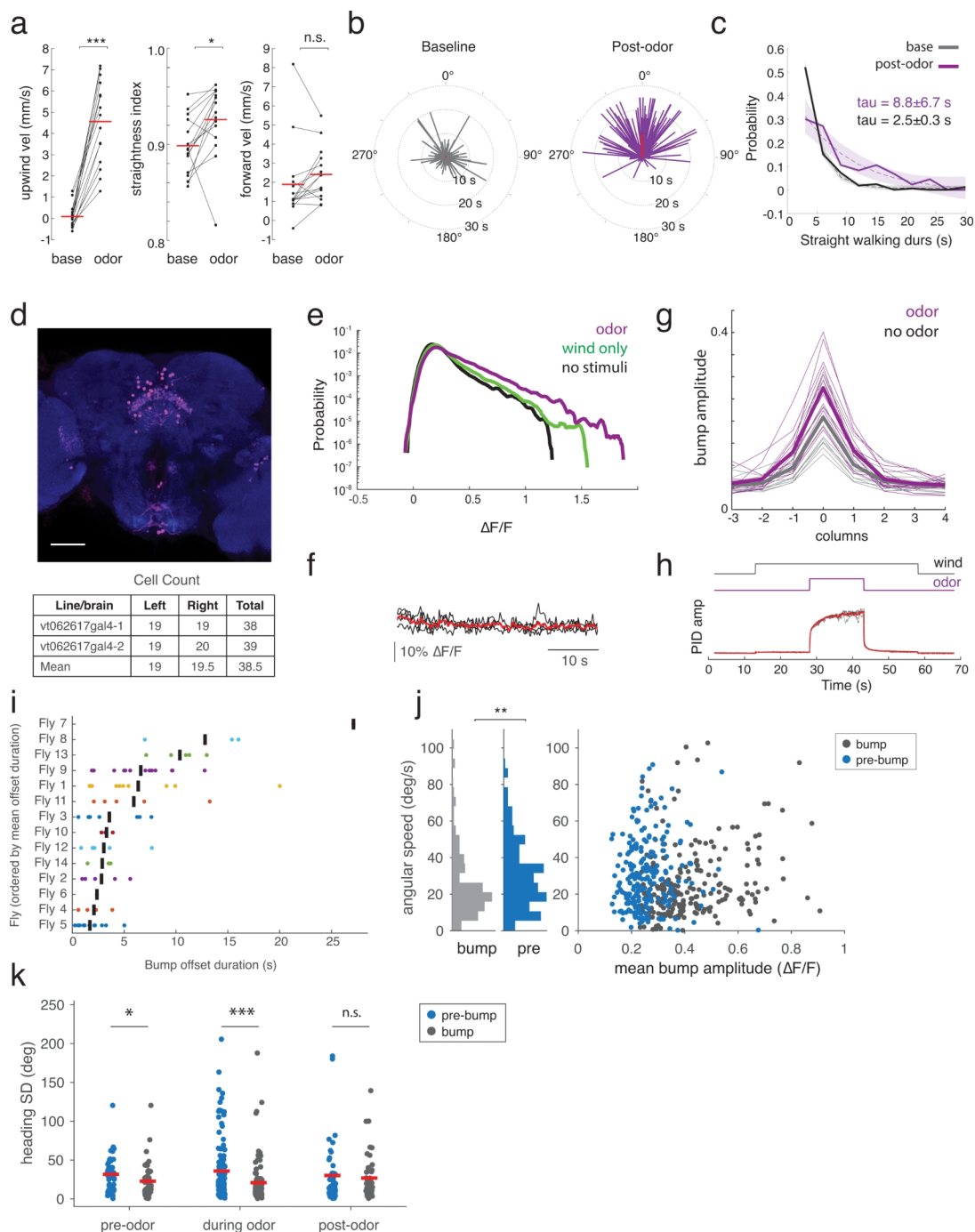


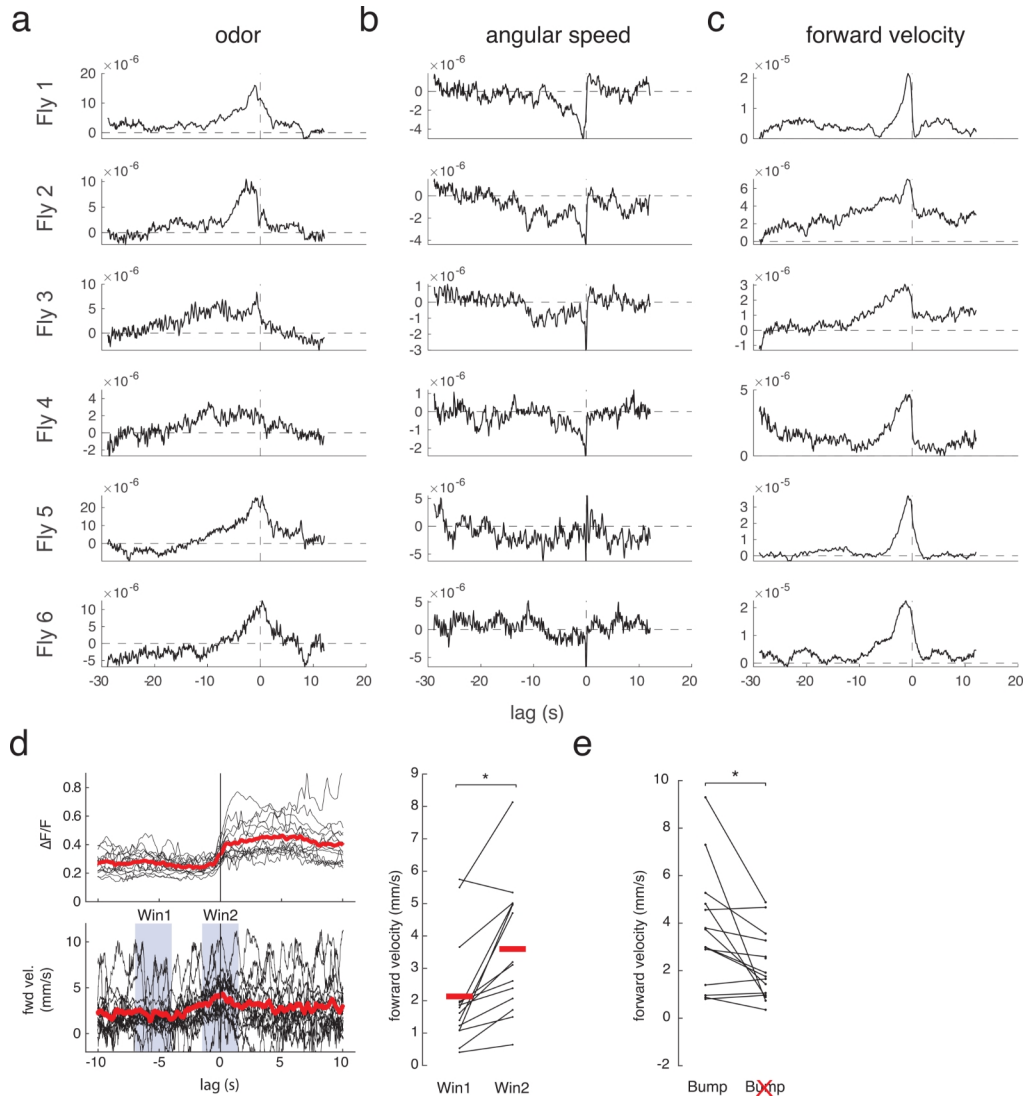
Supplementary Information



Supp. Fig. 1: Additional data on VT062617 FB local neuron responses.

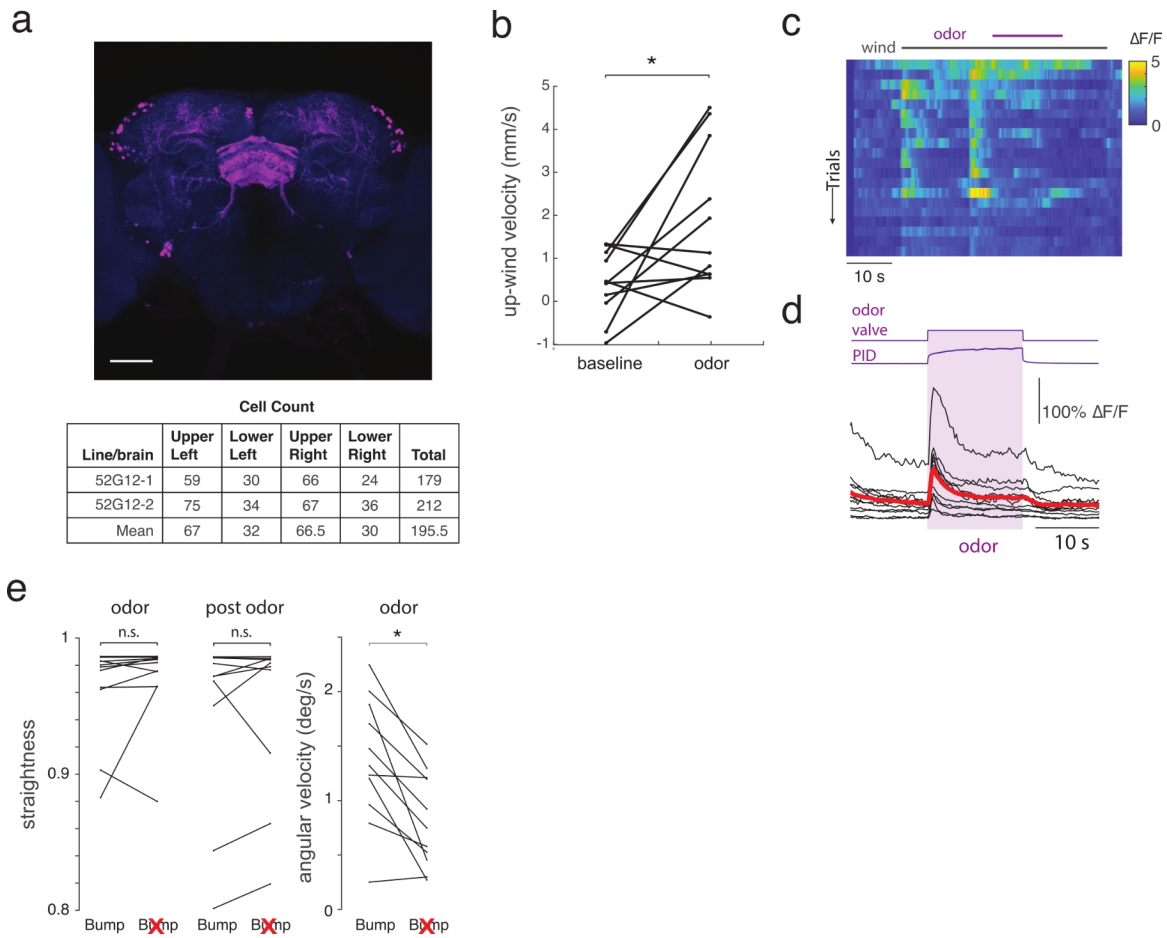
a: Characteristics of virtual trajectories during baseline versus odor periods for all flies (N=15). Each line represents the mean for one fly. Mean upwind velocity and straightness are significantly higher during odor than baseline (paired t-test, $p = 0.001$, $p = 0.035$). Forward velocity is not significantly different ($p = 0.43$). **b.** Heading distribution of persistent walking during baseline (left) and odor OFF (right) periods for all trials with bump activity ($n = 135$ trials from $N = 14$ flies). Mean resultant vector during post-odor (red) is biased upwind (mean: -1.5°). The length of each line represents the duration the animal walked straight after the respective time (odor: odor OFF, baseline: trial start). **c.** Distribution of persistent straight walking durations after odor (purple) and during baseline (grey) for each fly ($N = 14$). Exponential curves (dashed) with functional confidence intervals (shaded) were fit to the data, providing exponential decay values ($\tau_{\text{odor}} = 8.8 \pm 6.7$ s,

$\tau_{\text{base}} = 2.5 \pm 0.3\text{s}$). **d**: Top: Confocal image of VT062617>GCaMP brain, stained for GCaMP (pink) and nc82 (blue). Fluorescence signals were imaged from the dorsal (output) projections of FB local neurons. Scale bar is 50 μm . Bottom: table of cell body counts for left and right hemispheres in two brains. **e**: Distribution of $\Delta F/F$ values across all recordings in the presence of odor (purple), during wind only (green) and in the absence of stimuli (black). Total of 328 trials from 15 flies for each condition. **f**: Average bump amplitude in trials with no wind or odor stimuli, but motor still in closed-loop with heading (n=3-7 trials, N=3 flies). **g**: Mean bump profile across columns. Thin lines represent mean across trials (9 to 52 trials for each fly) and thick lines represent mean across flies (N=14 flies with both odor and no-odor imaging data). **h**: Photoionization detector (PID) recording of odor concentration during typical wind and odor pulse trials (n=6 trials, black: individual trials, red: mean across trials) at the location of the fly. Wind and odor control signals above. **i**: Mean bump persistence distributions of individual flies are significantly different from one another (one-way ANOVA, $p = 2.29\text{e-}10$, $n = 82$, $N = 14$). Each row shows all bump persistence times for a single fly, with the mean indicated by a black line. Flies are sorted along the y axis by mean duration. **j**: Mean angular speed (y-axis) versus bump amplitude (x-axis) for all bumps (222 bumps from 14 flies, blue) and for equivalent durations before each bump. Marginal distributions at left. Angular speed during bumps is significantly lower than during pre-bump periods ($p=0.01$). **k**: Standard deviation of unwrapped heading was smaller during bump versus pre-bump periods for bumps starting during the pre-odor (n=52 bumps, $p=0.03$), and odor (n=123 bumps, $p=3.6\text{e-}6$) phases, but not for bumps starting during the post-odor period (n=47 bumps, $p=0.60$). The smaller reduction observed for bumps starting in the post-odor period may reflect the fact that pre-bump periods occur during odor in this case. Source data are provided as a Source Data file.

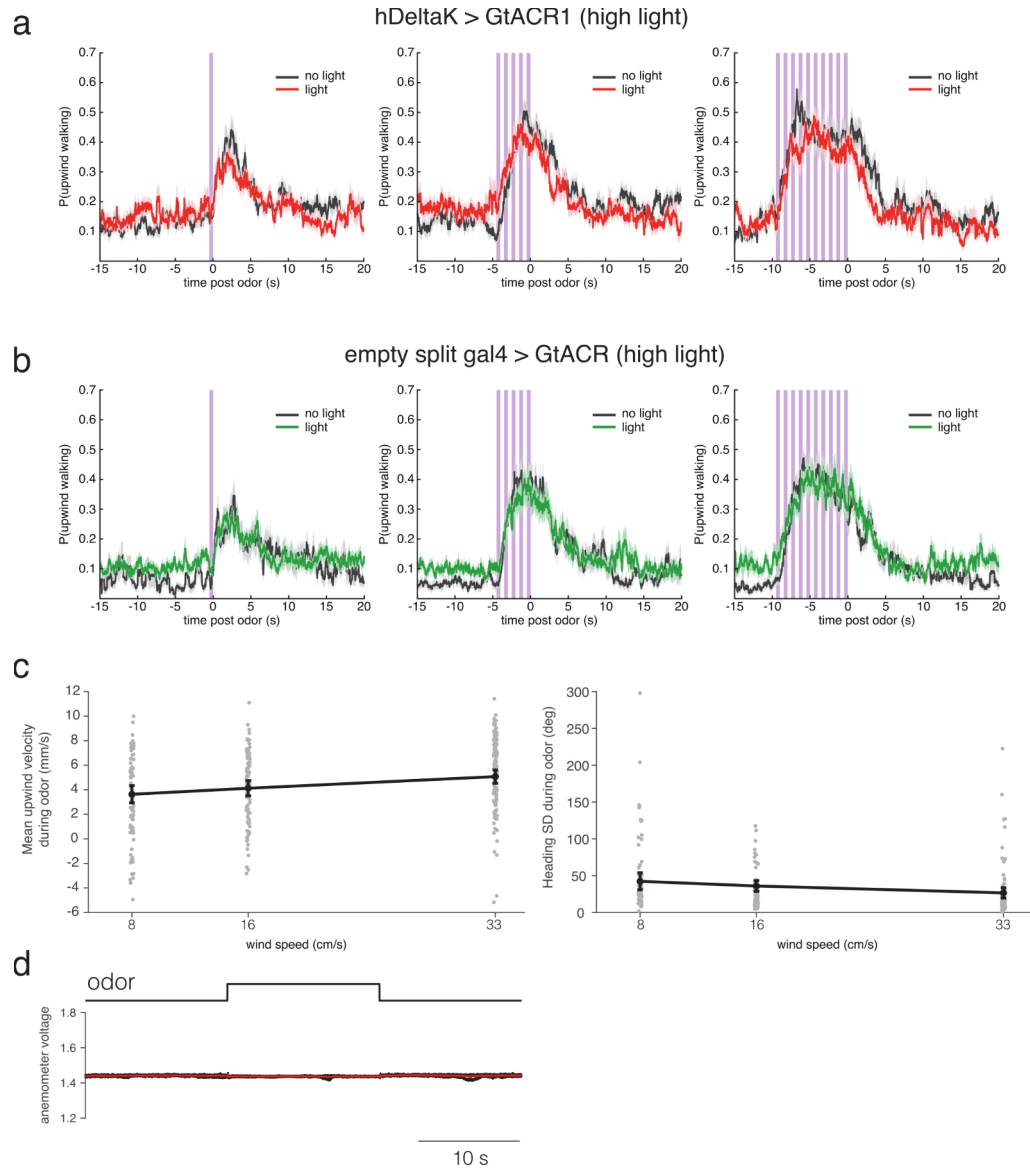


Supp. Fig. 2: Additional data on plume related activity and bump onset

a: Linear filters for odor (**a**), angular speed (**b**), and forward velocity (**c**) for each fly. **d:** Analysis of locomotor activity relative to bump onset in single odor pulse trials. Left: lower trace shows forward velocity aligned by bump onset (upper), which peaks at bump onset. Red traces represent means across flies (N=14 flies). Right: Mean forward velocity increases during bump onset (pre-onset mean=2.13, post-onset mean=3.60, paired t-test $p=0.046$, blue boxes indicate analyzed periods.) **e:** Mean forward velocity during odor is higher in samples with a bump than without (paired t-test, $p=0.032$, N=14). Source data are provided as a Source Data file.



Supp. Fig. 3: Additional data on responses of ventral local neurons labeled by 52G12. **a:** Confocal image of 52G12>GCaMP brains, stained for GCaMP (pink) and nc82 (blue). Scale bar is 50µm. Bottom: table of cell body counts for left/right/upper/lower quadrants of two brains imaged. **b:** Mean upwind velocities of flies expressing GCaMP in 52G12 neurons during odor versus baseline (N=11 flies, $p=0.018$). **c:** Mean activity in 52G12 neurons over time for all trials for one fly. Note transient responses to wind and odor onset. **d:** Time course of activity relative to measured odor time course (purple box) for all recorded flies. Black: mean for each fly (n=6 to 30 trials each), red: mean across flies (N=11 flies). **e.** Left: Mean straightness for 52G12 flies (N=11) in the odor and post-odor periods is not significantly different in the presence versus absence of a bump (paired t-test, during odor $p=0.966$, post-odor $p=0.893$). Right: Angular speed during odor is significantly higher in the presence of a bump (paired t-test, $p=0.029$). Source data are provided as a Source Data file.



Supp. Fig. 4: Additional data on silencing of hΔK neurons

a: Probability of upwind walking in hΔK split > GtACR flies responding to 1, 5, or 10 pulses of odor (500 ms at 1Hz) with light OFF (black, 1 pulse: n=287 trials from N=23 flies; 5 pulses: n=271 trials from N=24 flies; 10 pulses: n=297 trials from N=24 flies) or light ON (red, 1 pulse: n=362 trials from N=23 flies; 5 pulses: n=367 trials from N=24 flies; 10 pulses: n=322 trials from N=24 flies). Light level for these experiments was 76 $\mu\text{W}/\text{mm}^2$. We observed a decrease in upwind walking probability after odor OFF on light ON trials for the 10 pulse condition. **b:** Probability of upwind walking in empty-split > GtACR flies responding to the same stimuli with light OFF (black, 1 pulse: n=150 trials from N=26 flies; 5 pulses: n=234 trials from N=28 flies; 10 pulses: n=213 trials from N=29 flies) and light ON (green, 1 pulse: n=357 trials from N=26 flies; 5 pulses: n=324 trials from N=28 flies; 10 pulses: n=358 trials from N=29 flies). **c:** Dependence of upwind tracking on wind speed during walking ball trials. Left: upwind velocity during odor showed a positive correlation with wind speed ($\rho = 0.29$, $p = 3.2\text{e-}07$) but no significant relationship to wind speed by ANOVA ($p = 0.283$). Right: standard deviation of unwrapped heading during odor showed a negative correlation with wind speed ($\rho = -0.32$, $p = 2.4\text{e-}08$) but no significant relationship to wind speed by ANOVA ($p = 0.427$). Gray dots represent means of single trials; black circles represent mean across trials. **d:** anemometer trace (6 trials in black, mean in red) measured at the location of the fly during an odor pulse. Source data are provided as a Source Data file.

Supplementary Table 1: Reagents and Resources

Resource	Source	Identifier
FLY LINES		
VT062617-Gal4	VDRC	VDRC: 206875
52G12-Gal4	BDSC	RRID: BDSC_49581
Empty-Gal4	BDSC	RRID: BDSC_68384
w ¹¹¹⁸ 5905	Alvarez-Salvado et al. 2018	
UAS-GCaMP7f,UAS-tdTomato	BDSC/Nagel lab	RRID: BDSC_80906, 36327
w ¹¹¹⁸ ,s/CyO;UAS-GtACR1-EYFP/TM6b	Claude Desplan	
VT062617AD/[CyO,tb];VT037223DB/TM6b	Tanya Wolff	Janelia: SS63158
p65ADZp, ZpGAL4DBD (Empty Split-Gal4)	BDSC	RRID: BDSC_79603
SOFTWARE		
Python		Version 3.8
MATLAB	Mathworks	Version R2023a
Image J		Version 1.53t
		Version 2.0 (build date: Nov 2 2018)
FicTrac		
Pybmt	Mala Murthy lab	
CalmAn		
Register_roi2	Jason Moore	

Supplementary Table 2: Statistics for Figure 5. Significant p values are bolded.

Figure	Comparison	Sample size	Statistical test	p value
Fig. 5b-c	P(upwind orientation) for 1, 4, 7, or 10 odor pulses	N _{1 pulse} = 30 flies, 246 trials N _{4 pulses} = 28 flies, 226 trials N _{7 pulses} = 29 flies, 213 trials N _{10 pulses} = 30 flies, 246 trials	N/A	N/A
Fig. 5d	VT062617 > GtACR P(upwind orientation) for 1, 5, or 10 odor pulses with or without light	Light OFF: N _{1 pulse} = 30 flies, 381 trials N _{5 pulses} = 30 flies, 418 trials N _{10 pulses} = 30 flies, 463 trials Light ON: N _{1 pulse} = 30 flies, 405 trials N _{5 pulses} = 30 flies, 422 trials N _{10 pulses} = 30 flies, 450 trials	N/A	N/A
Fig. 5e	Empty-Gal4 > GtACR P(upwind orientation) for 1, 5, or 10 odor	Light OFF: N _{1 pulse} = 26 flies, 261 trials N _{5 pulses} = 26 flies, 246 trials	N/A	N/A

	pulses with or without light	trials $N_{10 \text{ pulses}} = 28 \text{ flies, } 318 \text{ trials}$ Light ON: $N_{1 \text{ pulse}} = 26 \text{ flies, } 294 \text{ trials}$ $N_{5 \text{ pulses}} = 28 \text{ flies, } 310 \text{ trials}$ $N_{10 \text{ pulses}} = 27 \text{ flies, } 292 \text{ trials}$		
Fig. 5f	Mean P(upwind orientation) for 1, 5, or 10 odor pulses with or without light, for VT062617 > GtACR and empty-Gal4 > GtACR1 flies	VT062617 > GtACR: $N_{1 \text{ pulse}} = 26 \text{ flies}$ $N_{5 \text{ pulses}} = 26 \text{ flies}$ $N_{10 \text{ pulses}} = 26 \text{ flies}$ empty-Gal4 > GtACR: $N_{1 \text{ pulse}} = 25 \text{ flies}$ $N_{5 \text{ pulses}} = 27 \text{ flies}$ $N_{10 \text{ pulses}} = 28 \text{ flies}$	Bonferroni-corrected two-tailed ranksum test ($\alpha = 0.017$)	VT062617 > GtACR 1 pulse: $p < 0.0001$ VT062617 > GtACR 5 pulses: $p = 0.0046$ VT062617 > GtACR 10 pulses: $p = 0.0004$ empty-Gal4 > GtACR 1 pulse: $p = 0.474$ empty-Gal4 > GtACR 5 pulses: $p = 0.046$ empty-Gal4 > GtACR 10 pulses: $p = 0.399$
Fig. 5g	Mean P(upwind orientation) for 1, 5, or 10 odor pulses with or without light, for hΔK split > GtACR and empty split-Gal4 > GtACR1 flies	hΔK split > GtACR: $N_{1 \text{ pulse}} = 23 \text{ flies}$ $N_{5 \text{ pulses}} = 24 \text{ flies}$ $N_{10 \text{ pulses}} = 24 \text{ flies}$ empty split-Gal4 > GtACR: $N_{1 \text{ pulse}} = 26 \text{ flies}$ $N_{5 \text{ pulses}} = 28 \text{ flies}$ $N_{10 \text{ pulses}} = 29 \text{ flies}$	Bonferroni-corrected two-tailed ranksum test ($\alpha = 0.017$)	hΔK split > GtACR 1 pulse: $p = 0.5399$ hΔK split > GtACR 5 pulses: $p = 0.065$ hΔK split > GtACR 10 pulses: $p = 0.0079$ empty split-Gal4 > GtACR 1 pulse: $p = 0.764$ empty split-Gal4 > GtACR 5 pulses: $p = 0.991$ empty split-Gal4 > GtACR 10 pulses: $p = 0.417$